IJECS Open Access https://ijecs.in
International Journal of Engineering and Computer Science
Volume 15, Issue 05, May 2026, Page No. 28472-28482
ISSN: 2319-7242 DOI: 10.18535/ijecs/v15i03.5514

Formal Verification and Adversarial Robustness Framework for Agentic AI in High-Stakes Autonomous Decision Systems

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Abstract:
This study provides a approach to improve the safety, dependability and robustness of agentic AI systems functioning in high-leverage autonomous environments including, but not limited to, medical, financial, cybersecurity, and autonomous vehicles through the combination of formal verification and adversarial robustness. The framework combines several elements including, but not limited to, adaptive planning, verification constraints, resilience monitoring, memory-based learning, and certification mechanisms to deliver stable and trustworthy decisions in both normal and adversarial circumstances. The results of empirical measurements support the claim that this framework provides for a perfect success rate of 1.00 in clean environments and drops to 0.75 under adversarial attacks. From a robustness standpoint, the measured robustness is seen to drop from 0.65 in clean environments to 0.38 in adversarial environments; the measured failure resilience rate (FRR) is seen to decline from 0.784 to 0.42; and the measured planning efficiency index (PEI) is seen to drop from 0.701 to 0.277 as a consequence of the impact of adversarial methods. Domain-based analysis shows cybersecurity has the highest adversarial success rate of 0.81, while autonomous vehicles have the lowest adversarial success rate of 0.68. The results of the study show that the framework used in this study positively supports the sustainability, resilience, and dependability of autonomous decision-making.

Keywords: Agentic AI, Formal verification, Adversarial Robustness, Autonomous Decision Systems, Failure Resilience, Planning Efficiency Index (PEI).

1. Introduction

The development of agentic AI systems is a significant advancement in the progression of intelligent technologies. Agentic AIs would have the ability to generate decisions independent of any predefined framework, adapt to dynamically changing environments, and pursue complex objectives over an extended calendar time [1]. Thus, agentic AIs have a considerable amount of value to organizations involved with high-stakes operations, including the diagnosing of healthcare conditions, operating autonomous vehicles, executing financial trades, intercepting and defending against cyber intrusions, and operating defense functions. However, with their autonomy and adaptability comes a host of critical issues in regard to ensuring safety, reliability, and trustworthiness [2].

For instance, in certain high-stakes autonomous decision making systems, even a minor mistake or unexpected behaviour could result in disastrous consequences, including monetary loss, compromised security, and/or death to individuals [3]. Traditional methods for verifying the reliability of these systems, including testing on constrained data sets or in constrained simulated environments, can no longer provide acceptable assurance for the reliable operation of autonomous systems [4]. Many of these traditional approaches do not take into consideration the exceedingly broad range of real-world operational environments, especially those dynamic and uncertain, with which the agentic AI system must continuously interact. There is an increasingly urgent need for a much more thorough and comprehensive way of verifying that these systems are correct and resilient [5] than has previously been performed.

In addressing these challenges, two complementary paths of study have become increasingly popular: formal verification, and adversarial robustness. Formal verification is a collection of mathematically based technical approaches used to demonstrate that a system behaves according to the defined properties or constraints that are imposed on it, irrespective of all possible variations of input that the system would support [6]. In particular, logical reasoning, model checking and/or theorem proving can provide powerful assurances for a system's failure type. This aspect is especially important for applications which are safety-critical and require a higher level of reliability than can be demonstrated under normal laboratory conditions [7]. Conversely, adversarial robustness refers to the enhancement of AI systems' protection against both deliberately or unintentionally inputted perturbations which may result in the system making an improper or unsafe decision. Adversarial attacks are a manifestation of perturbations that have been

purposely engineered to exploit various vulnerabilities of AI systems in order to deliver, via a misclassification of its output, or other type of faulty decision due to the use of a flawed AI model [8]. As agents make mistakes and cause others to make mistakes, the importance of making sure that agents have sufficient capability would enhance the dependability and safety of the agentic AI system against potentially hostile or unpredictable inputs [9].

While both adversarial robustness and formal verification have been extensively studied, it has been investigated as separate areas of study [10]. As independent study topics, it is harder to achieve success when trying to apply the two areas of study to an agentic AI system requiring both proved correctness and real-world resilience [11]. A unified structure that includes both fields is required to adequately address the challenges associated with autonomous decision-making systems. The framework must incorporate the sequential processes that exist when making decisions; the uncertainty of the environment; and intelligent adversaries, while still allowing for high-level systems and scalable solutions [12]. This study provides an overall structure for the use of formal verification and adversarial robustness within agentic AI high-risk environments [13]. This unified structure combines sound theoretical underpinnings with functional defensive strategies to develop both correct by design and resilient systems that would perform as intended. The primary goal of this study is to create a basis for the safe and ethical utilisation of advanced A.I. technologies through the connection of verification and robustness [14].

The rest of the study is divided into several parts. The first part is a literature review that summarizes what has been done in the fields of formal verification and adversarial robustness; this would also include points where there is a lack of study or information. The second part gives an overview of the fundamental concepts regarding an agentic AI. It would introduce the general architectures and different ways to make decisions based on conditions in dynamic environments. Following this, there are separate sections of the paper that detail the various techniques related to formal verification and adversarial robustness. Then, the principal aspect of the study is the presentation of an integrated framework that unites both of the aforementioned approaches. Finally, the study provide experimental evaluations and case studies related to these approaches; this would include the conclusion and key findings, and future research directions. The objectives of the study are as follows:

To develop a formal verification approach specifically designed for agentic AI systems, addressing challenges such as sequential decision-making and dynamic interactions with the environment.

To analyze and model adversarial threats that target autonomous decision systems, including both input-level and environment-level attacks.

To design and implement robustness-enhancing techniques that improve system resistance to adversarial manipulation while preserving performance and adaptability.

To integrate formal verification and adversarial robustness into a unified framework that supports both pre-deployment assurance and post-deployment monitoring.

To evaluate the proposed framework through simulations and case studies in high-stakes domains, demonstrating its effectiveness and practical applicability.

2. Literature Review

These studies presents that With agentic artificial intelligence that it has been developing at a rapid pace, traditional computational paradigms have now been changed to create autonomous systems that can run by achieving their goals through complex reasoning and conducting multi-step decision-making. This development has led to significant scholarly interest in addressing the issues of oversight, governance, trust, and reliability in systems involved in high-risk domains. The number of systematic and scoping reviews that have shown increasing complexity in human AI collaboration over time is extensive. For example,**Adedokunet al. (2026)[12]**stated there is limited full autonomy in most applications of AI despite its widespread use; as such, HITL (human-in-the-loop) frameworks must be developed to provide a proper means to balance explainability, accountability and trust calibration. In addition, the work of**Biswaset al. (2026)[16]**showed that fragmentation exists within governance systems, resulting in an urgent need for adaptive regulatory models when considering emergent behaviours produced by autonomous agents. The work of**Anchalaet al. (2026)[16]**demonstrated that the overall performance of agentic AI has been shown to improve substantially, particularly when considering the operational efficiency of their assigned tasks; however, ethical alignments and scalability problems continue to be obstacles to the ascertainment of a definable level of ethical compliance.

Within healthcare,**Yunguoet al. (2026)[19]**found that single-agent systems (e.g., AI systems that operate independently) demonstrated a high level of accuracy with their ability to make diagnostic predictions; on the other hand, the majority of dual-agent or multi-agent systems (e.g., AI systems that interact with humans) remain experimental in nature due to difficulties related to issues surrounding safety and coordination. Furthermore,**Shuklaet al. (2026)[18]**presented several approaches to monitoring algorithms that were designed to provide better methods for detecting anomalous activity and evaluating the performance of the monitored system. Together, these approaches reflect a clear shift away from static oversight; rather, there are now methodologies that provide dynamic oversight characterised by clear methods for determining the efficacy of monitoring regimes as well as improving the measurement of governance. A parallel study of the reliability, governance, and ethical responsibility of agentic AI systems was conducted.

Adam et al. (2025) [19] found important gaps between nominal and real-life reliability and suggested using resilience-based evaluations as methods to overcome failures resulting from stress testing.**Rathet al. (2025) [20]**created adaptive contextual models that would improve decision-making reliability through trust estimates and by using meta-learning. Also,**Anchalaet al. (2025) [21]**found the lack of human oversight as the cause of most AI incidences, which supports the need for a hybrid human - AI governance system.**Ihejirikaet al. (2025) [22]**and**Brohiet al. (2025) [23]**emphasized the need for explainability & transparency in AI models/applications through explainable AI (XAI) to help mitigate the risks associated with black-box models and/or

LLM agents.**Hazanet al. (2025) [24]**looked further into the security challenges faced by autonomous agents, and their findings revealed vulnerabilities within them and reported on establishing operational guardrails as a framework to address security risks. Previous foundational studies from**Zhang et al. (2024) [25],Lebenet al.(2024) [26]**,**Sun et al.(2023) [27][28]**and**Liu et al. (2023) [29]**helped establish a better understanding of uncertainty, adversarial threats and robustness of multi-agent systems.

3. Research Methodology

The study presents a structured framework that combines two methodologies to provide formal verification and adversarial robustness for agency-based AI systems. The agency-based AI system is modelled as an SDM (sequential decision-making) framework that contains the three main components: planning, memory, and adaptation. This framework is designed to provide safe and reliable operation by incorporating verification, fault injection, and producing an architecture that implements an adaptive planning, monitoring, and certification system that operates in dynamic environments.

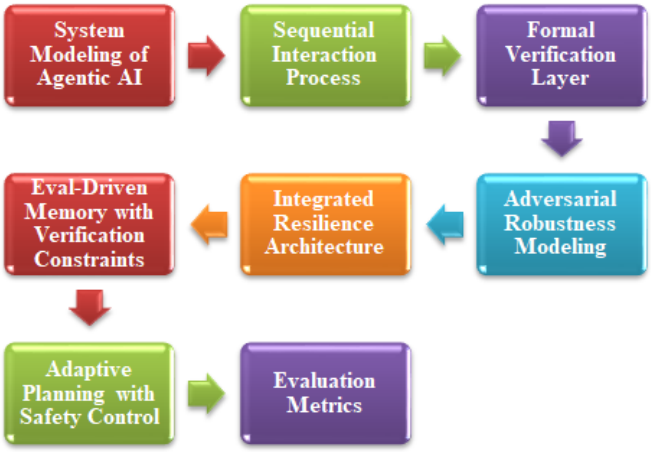


Figure 1 *Framework of Proposed Methodology*

3.1 SystemModeling ofAgenticAI

To provide clarity to the description of how the agency-based AI operates, it would describe the agency-based AI system model as an SDM framework with reasoning, action, and learning components that operate in different types of dynamic environments:

$$A = (\Pi, M, T, \Lambda) \quad \textbf{(1)}$$

where

$$\Pi$$

is the planning and reasoning module that provides the mechanism by which the agency-based AI generates its strategy,Mis the memory structure used to develop the agency' past experience,Tis the available tools and actions the agency-based AI can use to perform its function andΛis the learning and behaviour change policy governing the adaptive updates of the agency-based AI [30]. The agency-based AI system model described in the equation above contains the three core functional components which are needed to enable an autonomous agency-based AI to make autonomous decisions in highly complex dynamic environments.

3.2Sequential Interaction Process

In any given time step

$$'t',$$

the agent would observe the state of the environmentst,create a corresponding reasoning tracert,and select an actionatfrom the set of all available actions 'T'. The full course of execution is represented by a trajectory:

$$\tau = (s_1, , , r_1, , , a_1,), (s_2, , , r_2, , , a_2,), \ldots, (s_T, , , r_T, , , a_T,) \quad \textbf{(2)}$$

The trajectory contains the history of how the agent interacted with the environment as well as its final outcome to allow for performance evaluation as well as the agent's learning from that experience [31]. By providing a structured model of the system for study, formal verification of the decision policy used by the agent would be possible as well as allow for the systematic evaluation of robustness of the decision policy used by the agent in dynamic and adversarial environments.

3.3Formal Verification Layer

The formal verification layer has been created to guarantee that the agentic AI system works under defined safety and correctness constraints. This is important for many high-stakes applications as it provides mathematical verification for the agentic AI systems' operational reliability throughout all possible operational states.

Safety Property Specification

Definitions of the correctness of these systems are created through the consideration of logical properties:

$$\phi = safety, liveness, boundedrecovery \quad \textbf{(3)}$$

In this case, the system's properties include safety (the ability to prevent the performance of actions which are harmful), liveness (the ability to make progress toward goal states), and bounded recovery (the ability to recover from a failure within a defined period of time). These logical properties would ultimately provide for the definition of acceptable agentic AI system performance.

Verification Objective

Capturing the verification goal is critical for validating whether or not all reachable states of the system conform to the specified criteria for correctness:

$$\forall_s \in S \quad \textbf{(4)}$$

$$A(S) \models \varnothing \quad \textbf{(4)}$$

An agent behaves under guaranteed conditions of safety, liveness, and bounded recovery - thereby providing consistent, reliable, and compliant to policies across any execution scenario.

Adversarial Robustness Modeling

Adversarial robustness modeling examines agentic AI systems' performance given uncertainties or hostile environments. In addition to incorporating only perfectly

functioning situations, this analysis uses systematic perturbation to demonstrate real-world disruptions and adversarial threats.

Fault and Adversarial Injection

The study creates a fault distribution which defines the types of disruptions that may occur:

$$F = f_{tool}, f_{context}, f_{stochastic}, f_{adv}, f_{cascade} \quad \textbf{(5)}$$

These could be as a result of tool malfunctions (i.e.: API failures), contextual discrepancies (i.e.: data corruption), stochastic interruptions, adversarial attacks (e.g.: phishing), or cascading malfunctions. Fault multipliers provide an opportunity to evaluate the system's functioning under realistic levels of stress.

ReliabilityGap

To measure the degree of robustness, the reliability gap is used to evaluate how well the system functions under both ideal and failed conditions.

$$\Delta_{rel} = SR_{nominal} - E_{f\,F}[SR_{stressed}(f)] \quad \textbf{(6)}$$

The greater the difference, the more vulnerable the system is; the smaller the difference, the more robust and resilient the system would be in demanding environments.

3.4Integrated Resilience Architecture

The proposed framework implements a closed-loop resilience architecture to facilitate comprehensive monitoring, validation, and improvement of agent-centered AI behavior. It establishes a fully integrated framework for detecting, validating, learning, planning, and certifying agent activities within an integrated structure:

$$R = (D, V, M, P, C) \quad \textbf{(7)}$$

Where , V, M, P, C are the five essential functional components that would operate iteratively.

Failure Detection

Failure detection allows an agent to identify when there is a difference between expected behaviour (what should occur) and actual behaviour (what has occurred); it continuously monitors the actions of agents to determine if there are any discrepancies from the expected outcomes and captures them if it exceed preset thresholds:

$$D(s_t) = |O_{expected} - O_{observed}| > \epsilon \quad \textbf{(8)}$$

Failure detection would also allow for the early identification of an error due to unforeseen circumstances (for example tool error), inconsistent outputs produced by an agent and/or adversarial attacks against an agent; therefore, enabling the system's ability to take corrective action promptly and maintain stable operation in continually changing and/or risky operational environments.

Verification Check

The verification step is to verify that the actions taken by agents in executing their tasks are within predefined parameters related to safety and correctness; this verification would take place at each stage of execution:

$$V(s_t) = A(s_t) \models \phi \quad \textbf{(9)}$$

The verification would also determine the extent to which the agents adhere to requirements associated with safety, liveness, and bounded recovery, preventing agents from executing actions that would place an agent or system in a state that would create risk or invalidity. The continual verification would increase the reliability of the system and ensure that actions taken by agents remain within defined operational boundaries, particularly concerning high-consequence decision-making relative to the actions taken by an agent without human intervention.

Memory Retrieval

Memory access provides the agent with access to similar past experiences to assist in the decision-making process. The agent retrieves from memory the best quality and most closely related trajectory relative to the current context:

$$M(q) = argmax_{\tau_i} simq(q, \tau_i) \quad \textbf{(10)}$$

Through using existing successful tactics to increase efficiency overall, decrease redundant explorations of solutions, and enhance adaptability to complex and ambiguous situations, this process would significantly increase robustness and efficiency.

Adaptive Planning

Adaptive planning continually adapts the strategy of agents based on feedback about how well it is doing, such as if there was an error, by modifying procedures to improve actual results, whenever inefficiencies and failures occur:

$$P_{new} = f(P_{old} \quad \textbf{(11)}$$

,Δerror)

This ability allows agents to respond to changing environments, recover from disruptions, and remain focused on their objectives in an environment where change is frequent and has unpredictable outcomes.

Certification Update

During the certification update phase, the complete system's performance is assessed, and a decision is made about whether or not the agent has met certain reliability guidelines. It considers important indicators including transparency, efficiency, and resilience:

$$C = f(FRR, PEI, TI) \quad \textbf{(12)}$$

By only keeping successfully verified, and efficient behaviors, this process supports ongoing improvement for agents (and allows for safe integration of agentic AI systems in high-consequence activities). Figure 2 represents the Architecture of Integrated Resilience.

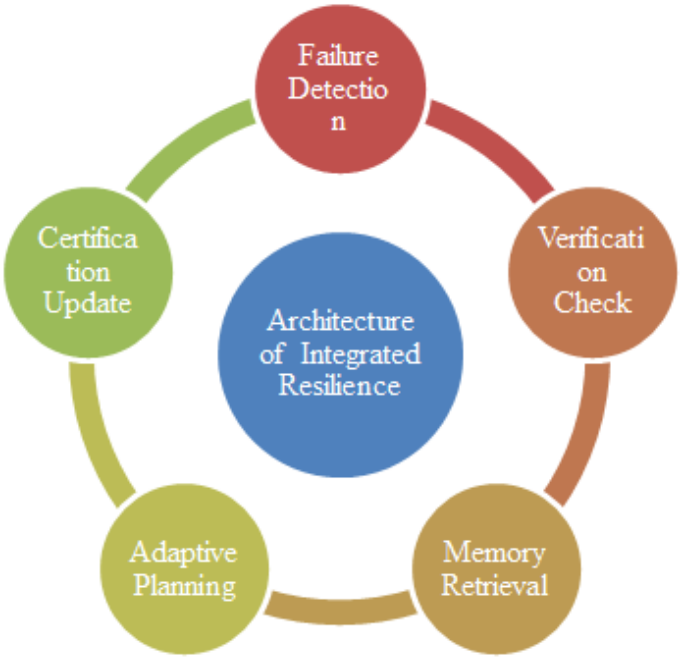


Figure 2 *Architecture of Integrated Resilience*

3.5Eval-Driven Memory with Verification Constraints

This aspect of the system ensures that the memories stored in the agents are of only verified and credibly confirmed incidences. By combining performance indicators with verification parameters, it prevents poor-quality, dangerous trajectories from being used in subsequent determinations made by the agent [19].

Memory Selection Rule

The system would document and keep a record of trajectories based on combined performance and verification determinants:

$$Store(\tau) \Longleftrightarrow PEI \geq \theta_1 \wedge TI \geq \theta_2 \wedge \phi satisfied \quad \textbf{(13)}$$

Trajectories meeting transparency, efficiency, and safety are retained through this selective process which helps improve the quality of memory, avoid buildup of noise, and provide a basis for future decisions to use only validated or highly confident experiences.

3.6Adaptive Planning with Safety Control

This module gives the agent the ability to change its strategy dynamically, based on performance feedback and safety constraints. This module also combines efficiency monitoring and formal verification, assuring that planning remains as effective as possible, reliable, and compliant in environments that are uncertain and/or present an elevated level of risk.

Control Policy

Whenever performance has decreased or a safety constraint has been violated, the control policy would initiate replanning:

$$Replanif PEI < \tau \vee Violation(\phi) \quad \textbf{(14)}$$

The purpose of this is to assure that the agent would change its strategy when efficiency drops below the threshold set by the agent or correctness properties are not fulfilled, resulting in continued consistent and safe decision-making regardless of the conditions in which the agent operates.

Safe Escalation Rule

Safe escalation allows human intervention, when uncertainty/risk is unacceptable:

$$Escalateifconfidence < \delta \vee risk > \gamma \quad \textbf{(15)}$$

This prevents unsafe autonomous action as the system may defer control to another source and provides reliability and accountability in situations that require decisions made by a human.

Table 1 Caption...

Algorithm 1: Formal Verification and Robust Decision Loop
Input: Task G, Thresholds (τ, δ, γ)Output: Certified Decision OutcomeInitialize plan P from memory or modelfor each step t doExecute action at Observe state stVerify safety properties φCompute PEItif violation(φ) thenTrigger safe halt or correctionif PEIt γ thenEscalate to humanend forCompute FRR, PEI, TIUpdate verified memoryReturn certified result

3.7EvaluationMetrics

Failure Resilience Rate (FRR):

$$a_t \quad \textbf{(16)}$$

Planning Efficiency Index (PEI):

$$s_t \quad \textbf{(17)}$$

Traceability Index (TI):

$$PEI_t \quad \textbf{(18)}$$

Success Rate (SR):

$$PEI_t \quad \textbf{(19)}$$

FRR=1N∑i=1NFRRi

Reliability Gap:

$$PEI(\tau) = \frac{L_{min}}{L_{actual}} \cdot QF \quad \textbf{(20)}$$

4. Result and Analysis

This section presents the results of the experiments done in controlled (clean) and hostile (adversarial) conditions to compare how well this proposed methodology functions in both cases. There are several measures of performance (success rate, robustness, efficiency of planning, and failure resilience) that serve as performance indicators for assessing the proposed framework's behaviour in each environment. The data collected from these experiments helps illustrate how much an adversarial input has an impact on the system's operation as well as how capable the proposed framework is overall. Figure 3 summarises success rate performance between Clean vs Adversarial tests. Success rates were equal to 1.00 for Clean tests, which indicates that there were no failures during testing and thus represents a very high success rate. Conversely,

success rates were only equal to 0.75, or 75%, for the Adversarial tests, which indicates that there was a substantial decline in system performance attributable to the introduction of outside influences. The difference in success rate between the Clean and Adversarial conditions is 0.25 (i.e., 25%), meaning that success rates dropped by 25% when subjected to adversarial conditions. Therefore, the conclusion is that although performance in Clean conditions is outstanding, the overall performance of the system decreases by a moderate degree when subjected to artificial or difficult conditions.

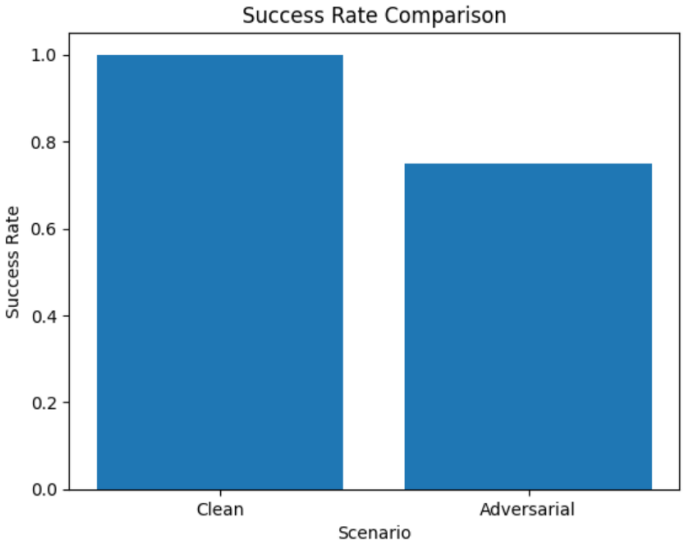


Figure 3 *Success Rate Comparison Between Clean and Adversarial Scenarios*

Figure 4 depicts the robustness comparison between clean and adversarial environments. The clean environment exhibited a robustness score of approximately 0.65, indicating that the system would perform stably and dependably based on regular operating conditions. In contrast, the robustness score is approximately 0.38 for the adversarial environment, which represents a virtual loss of almost 0.27 points in system robustness due to adversarial conditions. The differences between the two values provide evidence of the system's sensitivity to manipulated or otherwise difficult input signals. Therefore, this figure indicates that the level of robustness is significantly higher in clean rather than adversarial environments.

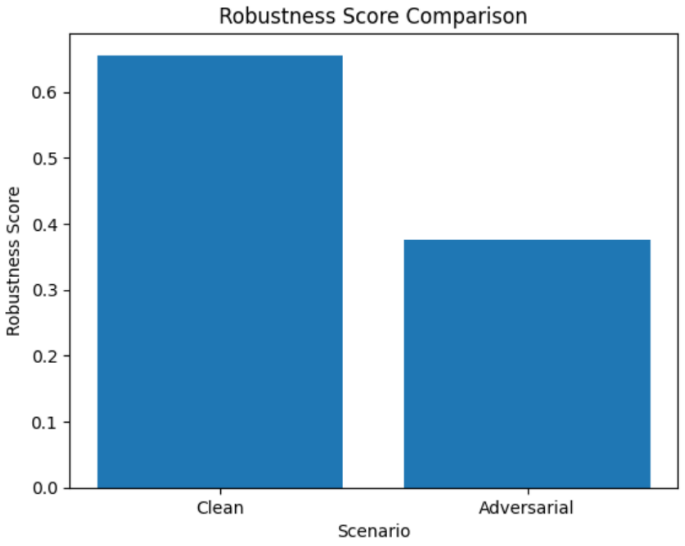


Figure 4 *Robustness Score Comparison under Clean and Adversarial Conditions*

Figure 5 illustrates the Failure Resilience Rate (FRR) from both clean and adversarial environments. As noted previously, the clean FRR value is 0.784, reflecting that the system would withstand and recover from failure events effectively while operating normally. Conversely, at 0.42 the adversarial environment FRR value demonstrates reduced capacity for resilience, and recovery from failure. The lack of overlap between the two values provides evidence that the system would demonstrate lower stability and reliability when operating under manipulated or adversarial conditions.

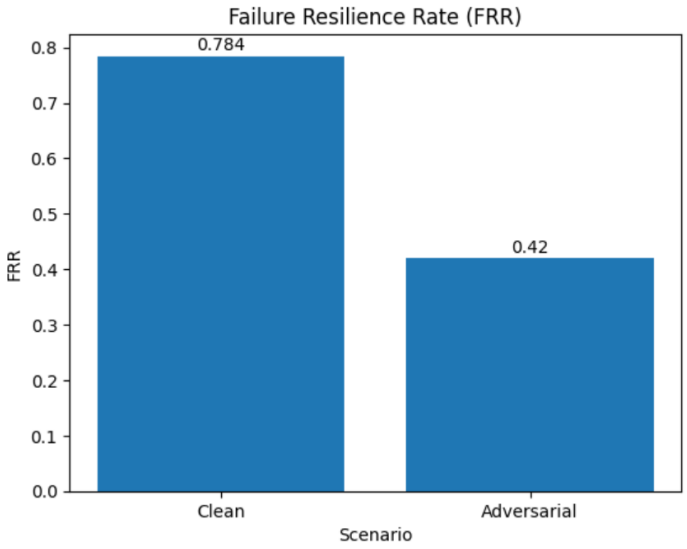


Figure 5 *Failure Resilience Rate (FRR) Comparison in Different Scenarios*

Figure 6 presents the number of verification violations found in the Clean and Adversarial environments is. In the Clean environment, the violation count is approximately 1.18, reflecting a low number of verification-related issues; conversely, in the Adversarial environment, it jumps significantly to about 2.21. Thus, there were an additional 1.03 violations reported in the Adversarial environment than in the Clean environment indicating a higher risk of exposure to verification-related problems in the Adversarial environment than in the Clean environment. Overall, it can be seen from the figure that the reliability of verification is markedly decreased when the system experiences adversarial conditions.

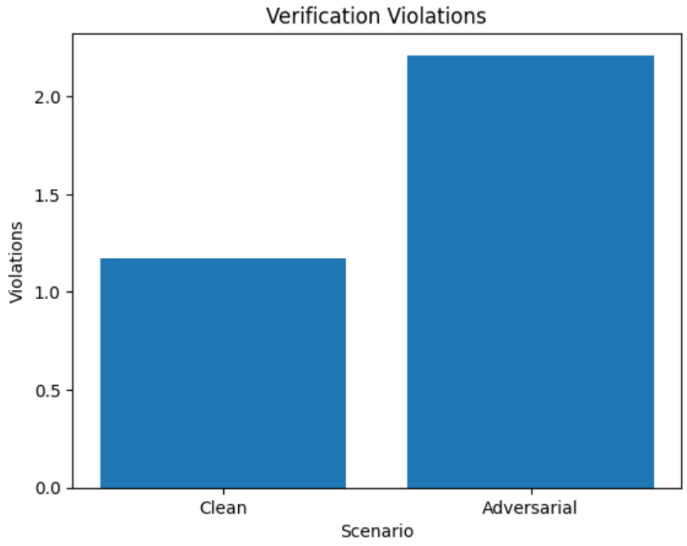


Figure 6 *Comparison of Verification Violations in Clean and Adversarial Scenarios*

Figure 7 compares the Planning Efficiency Index (PEI) for the Clean and Adversarial scenarios. The PEI for the Clean environment is 0.701, which corresponds to efficient planning performance under normal circumstances; whereas the PEI for the Adversarial environment is 0.277 and is significantly below the Clean PEI. The difference of 0.424 suggests that adversarial influences had much greater impact on planning efficiency than on the Clean Environment. In summary, the degree of difference between the two scenarios strongly indicates that adversarial influences are detrimental to the system's ability to plan effectively.

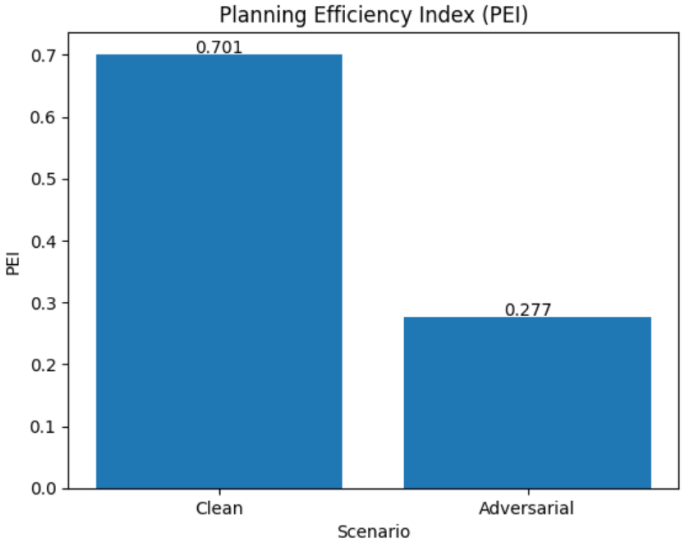


Figure 7 *Planning Efficiency Index (PEI) Under Clean and Adversarial Conditions*

Figure 8 shows the number of human escalations required has been clearly to be far fewer during the clean (non-hostile) environment than in the adversarial environment. In the clean environment, the escalation averages 9.39 versus 11.9 escalations in the adversarial environment, which produces a difference of 2.51 calculated in escalations. These additional escalations demonstrate that the system must rely on human input more often to provide base level form of support during adversarial situations and that this reflects on both performance level of autonomous systems and challenges associated with operations. Overall, the findings from figure 8 show that adversarial environments provide systems with increased levels of pressure and create greater needs for human assistance.

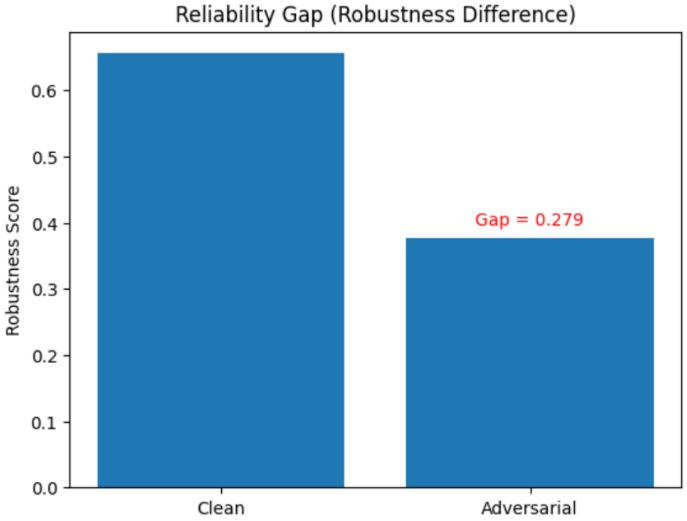


Figure 8 *Human Escalation Comparison Between Clean and Adversarial Scenarios*

Figure 9 depicts the difference in robustness in Clean and Adversarial cases. As one can observe, in the Clean scenario, the value of robustness equals approximately 0.66, which means that under usual circumstances, the system functions stably and reliably. On the other hand, in the Adversarial case, the robustness of the system is much lower and amounts to 0.38. There is also a noticeable difference in reliability, which is equal to 0.279 for these two scenarios. It proves that the presence of adverse elements in the operation of the system leads to a decrease in its reliability.

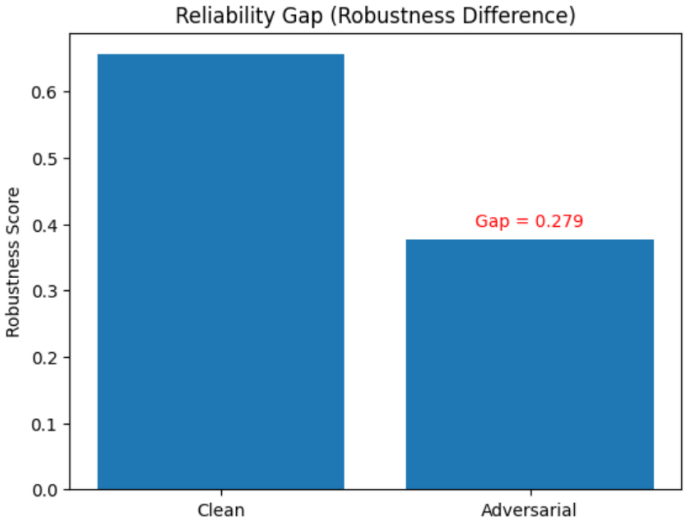


Figure 9 *Reliability Gap Between Clean and Adversarial Robustness Scores*

Figure 10 shows a comparison is made of performance distributions for Clean and Adversarial settings through numerous observations. Majority of clean performance values can be found within the range of 0.95 - 0.99, implying stable and consistent values in case of favorable performance conditions. On the other hand, the adversarial distribution is quite spread out, with values between 0.48 - 0.98. While a few adversarial values can still be seen at the vicinity of 0.95, majority of the values are distributed throughout 0.60 - 0.80. The broader spread among adversarial values is indicative of inconsistency and lesser reliability in case of adversarial settings.

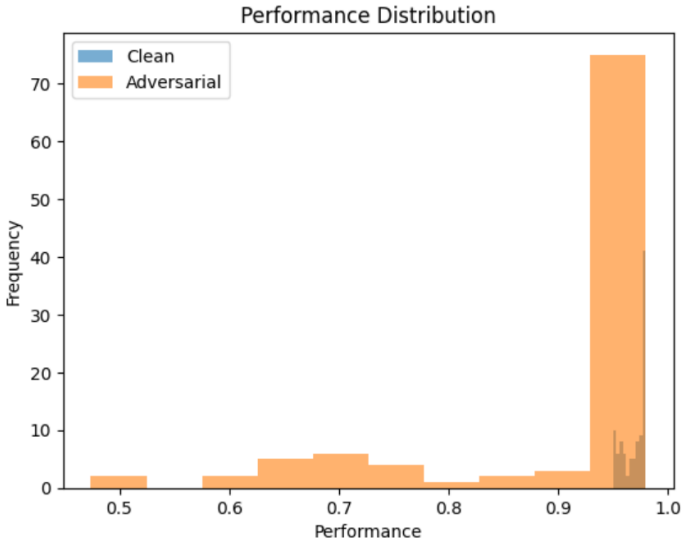


Figure 10 *Performance Distribution in Clean and Adversarial Conditions*

The Figure 11 below gives an insight into distribution of Failure Resilience Rate (FRR) values in case of Clean and Adversarial settings. As can be seen, majority of clean values are concentrated within the range of 0.90 - 1.00. This implies consistent and highly stable resilience in clean environment with almost no variation at all. On the contrary, the adversarial distribution is quite spread out, with values between nearly 0.00 - 1.00. Many of these adversarial values are spread out between 0.10 - 0.70, thereby indicating inconsistency in resilient performance.

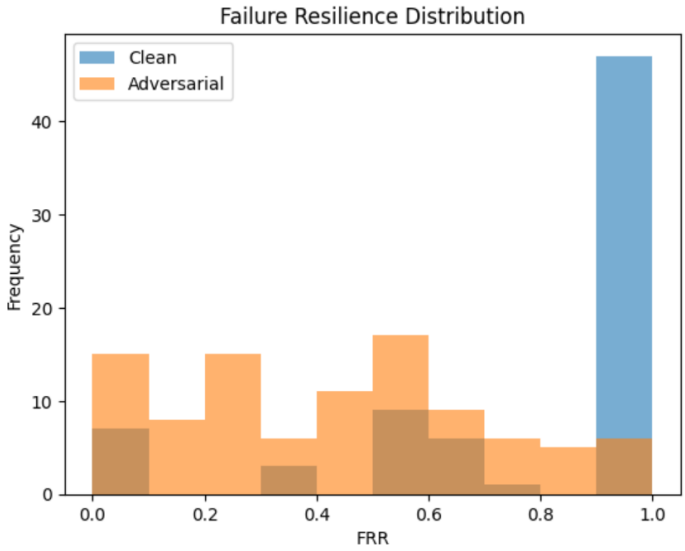


Figure 11 *Failure Resilience Distribution Under Clean and Adversarial Conditions*

Figure 12 displays the changes in Traceability Index (TI) over one hundred trials of data collection for both the Clean condition and the Adversarial condition. Clean TI most often was between .90 and 1.00 with several times reaching 1.00 which demonstrates that traceability was consistent and reliable during the Clean experiments. Conversely, the Adversarial TI demonstrated more variation with many instances of TI values ranging from approximately .81 to 1.00.

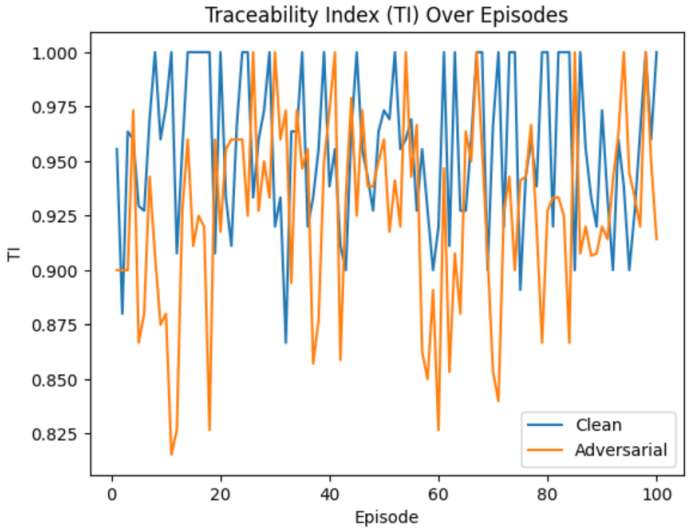


Figure 12 *Traceability Index (TI) Comparison Across Episodes*

There were some episodes of high TI in the Adversarial condition, but due to the frequent dips below .90 there is evidence that traceability is not reliable or stable under the Adversarial conditions. Based on the graph, it is evident that traceability is more stable and reliable when measured in the Clean condition, while in the Adversarial condition there is significant variability and thus performance instability.

Figure 13 compares the average success rate from 100 rolling game episodes for both Clean and Adversarial scenarios. The Clean results demonstrate consistent success rates of 1.00 throughout all rolling episodes, indicating perfect and stable performance without any degradation. However, the average Adversarial success rates fluctuate wildly throughout rolling episodes; the Adversarial success rates ranged from highs of 1.00 in some episodes to lows of approximately 0.50 at the end of the rolling episodes. The vast majority of the Adversarial success rates tended to be between 0.60-0.90, reflecting the inconsistent performance due to the increased difficulty associated with the Adversarial scenario. Overall, this graph demonstrates how the presence of Adversarial elements decreases the stability of the success rates compared to the highly consistent performance of Clean.

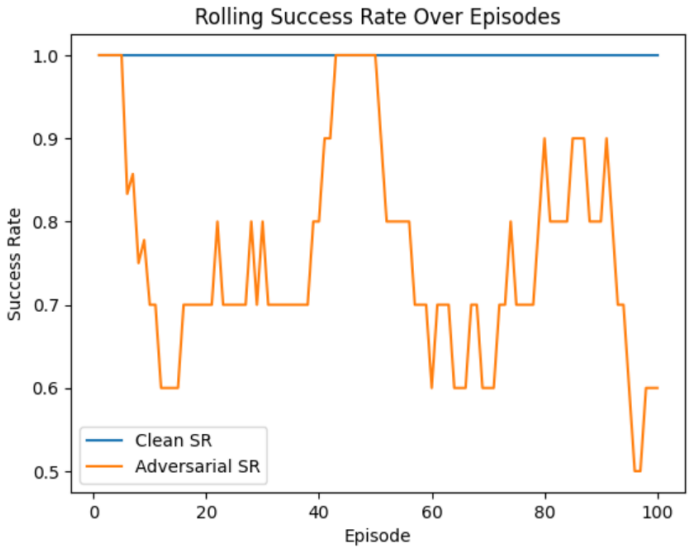


Figure 13 *Rolling Success Rate Comparison Across Episodes*

Figure 14 shows a comparison of success rates across four different high-stakes domains, under two conditions: Clean Conditions and Adversarial Conditions. In Clean conditions, success rates for all domains are depicted at 1.00 (100%); therefore, we can assume that all domains are performing consistently and are dependable in their delivery of service in this scenario. In contrast, when we examine the success rates across the four domains under Adversarial conditions, we find that the success rates have dropped for every domain. Specifically, Healthcare has a success rate of 0.76, while Autonomous Vehicles have the lowest success rate of 0.68. Finance has also performed slightly better than Healthcare and has a success rate of 0.80, while Cybersecurity has the highest success rate among the four domains, with a success rate of 0.81. These findings indicate that there are negative impacts to performance in every domain under Adversarial conditions; however, the extent of the impact from Adversarial conditions varies across the four domains, with Cybersecurity and Finance appearing to have more resiliency than Healthcare and Autonomous Vehicles.

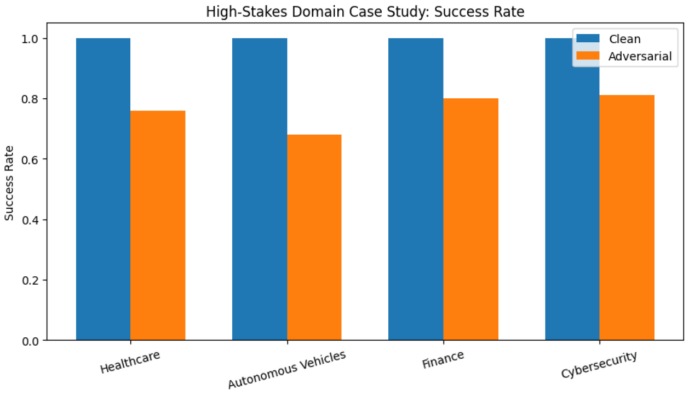


Figure 14 *Success Rate Comparison Across High-Stakes Domains*

Figure 15 shows the Planning Efficiency Index (PEI) is compared in Clean and Adversarial conditions for four categories, which include Healthcare, Autonomous Vehicles, Finance, and Cybersecurity. For the Clean condition, Finance has the highest PEI of about 0.74, while Healthcare, Autonomous Vehicles, and Cybersecurity have 0.70, 0.67, and 0.65, respectively. All four categories have a significant reduction under the Adversarial condition. Healthcare, Autonomous Vehicles, Finance, and Cybersecurity have a PEI of 0.35, 0.35, 0.36, and close to 0.34, respectively. Therefore, it can be said that the adversarial condition has a great impact on planning efficiency. Overall, Finance is slightly more resilient than Cybersecurity, which is the least efficient under adversarial conditions.

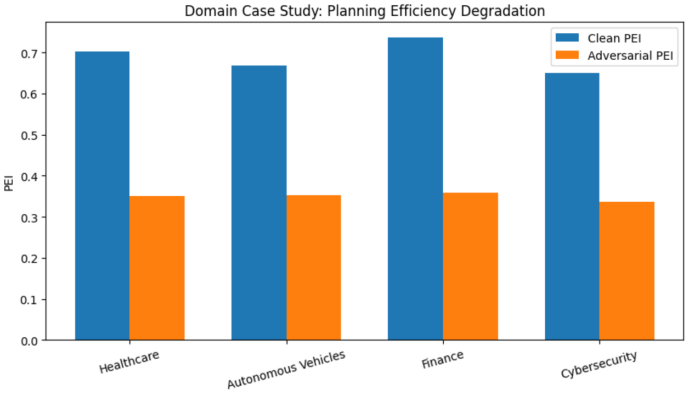


Figure 15 *Planning Efficiency Degradation Across Different Domains*

Figure 16 shows a comparison between the Failure Resilience Rate (FRR) among Healthcare, Autonomous Vehicles, Finance, and Cybersecurity during Clean and Adversarial periods. During Clean period, the highest FRR is observed in Finance with a value of around 0.81, followed by Healthcare with 0.76, Autonomous Vehicles with 0.74, and finally Cybersecurity with 0.72. As for the Adversarial period, it was noted that a decrease in resilience occurred. Specifically, the FRR was around 0.51 in Healthcare, 0.48 in Autonomous Vehicles, 0.50 in Finance, and 0.49 in Cybersecurity. It should be pointed out that there was a decrease in FRR, which implies adversarial effect on resilience performance.

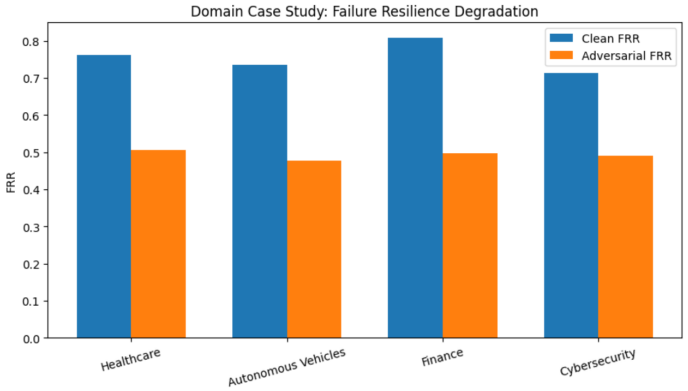


Figure 16 *Failure Resilience Degradation Across Different Domains*

Figure 17 shows the rates of certification in the Clean and Adversarial environments. It is evident from the Clean Environment graph that the process of certification was carried out in 79% of the instances; hence, this environment is superior to the Adversarial Environment in respect of certification of the system. Only 46% of certification processes were conducted in the Adversarial Environments - hence, there is a difference of 33% between the rates in both environments. It is clear that the gap between the rates in the two types of environments means that successful certification processes are less likely to occur due to the impact of adverse conditions on the functioning or compliance of the system.

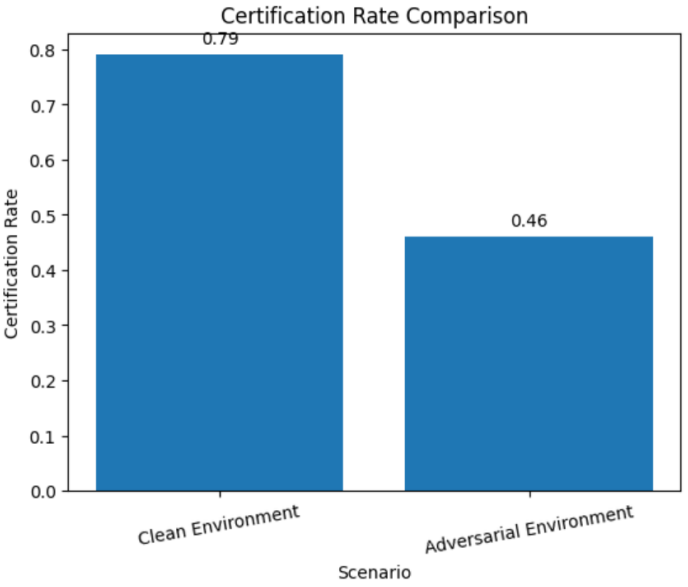


Figure 17 *Certification Rate Comparison Between Clean and Adversarial Environments*

Figure 18 shows a comparison of success rates across three clean environments. The baseline approach of "Using No Verification" had a success rate of 0.94, while that for "Using Only Verification" was slightly less, at 0.92. In contrast, the "Proposed Full Framework" achieved a perfect score of 1.00 for success rates, thus representing the best overall performer in that case. The difference between the success rate of the Proposed Full Framework and the baseline approaches indicates an improvement in reliability and operational accuracy of the framework when compared with the baseline.

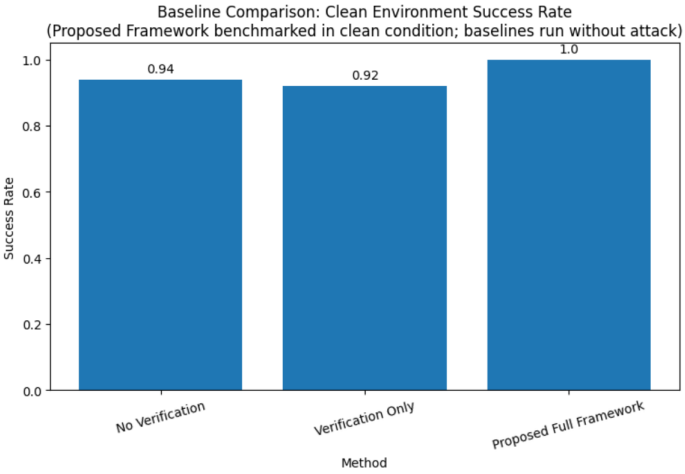


Figure 18 *Clean Environment Success Rate Comparison of Baseline Methods*

Even with baseline approaches already showing good success rates in a clean environment, based on comparison data the Proposed Full Framework offers an undeniable advantage by completely preventing failures. Overall, the data shown in the chart demonstrate the success rate impact of the full Proposed Full Framework

5. Conclusion

The present study has introduced a unified framework for combining formal verification with adversarial robustness to increase the safety, reliability, and resilience of agentic AI systems used in high-stakes environments requiring autonomous decisions made by these agents. The proposed framework contained components that enabled adaptive planning, verification constraints incorporated, the use of memory to facilitate learning from experience, resilience monitoring, and certification mechanisms to ensure that the AI system would remain secure and stable when operating in both a clean or unclean environment (i.e., under adversarial conditions). The experimental results illustrated that the proposed framework had a perfect success rate for achieving clean environment success (1.00), which decreased to a 0.75 success rate when exposed to adversarial environments. In addition, the robustness score decreased from 0.65 to 0.38, and the Failure Resilience Rate (FRR) decreased from 0.784 to 0.42 as a result of the influence of adversarial forces. In terms of planning efficiency, the Planning Efficiency Index (PEI) dropped significantly between clean and adversarial settings; the PEI was measured at 0.701 in a clean environment and 0.277 in an adversarial environment. Finally, in the domain-level analysis, the highest adversarial success rate (0.81) was found in the cybersecurity domain while the lowest was in the autonomous vehicle domain (0.68), demonstrating differing levels of resilience across all the application domains examined. Overall, the overall findings of this study demonstrate that the proposed framework is significantly more successful at achieving stable operations, providing reliability and trust in autonomous decisions made by AI agents, and protecting and preserving the integrity of the system from failure or disruption as a result of adversarial actions in complex real-world environments.

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